Between order and disorder: phase transitions in living systems

Statistical Physics of Living Systems

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Emergence – local interactions may lead to long range of order

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Biological intuition

Robustness

- Ability to maintain stable functions despite environmental fluctuations

Adaptability

- The ability to rapidly respond to stimuli

Eg. birds – must be disordered enough to feed effectively, but be able to organise into a group to protect themselves from predators. To understand how birds transition between states, we first need a mathematical way to quantify *order*.

Munoz, 2018

Classical phase separation – order parameters

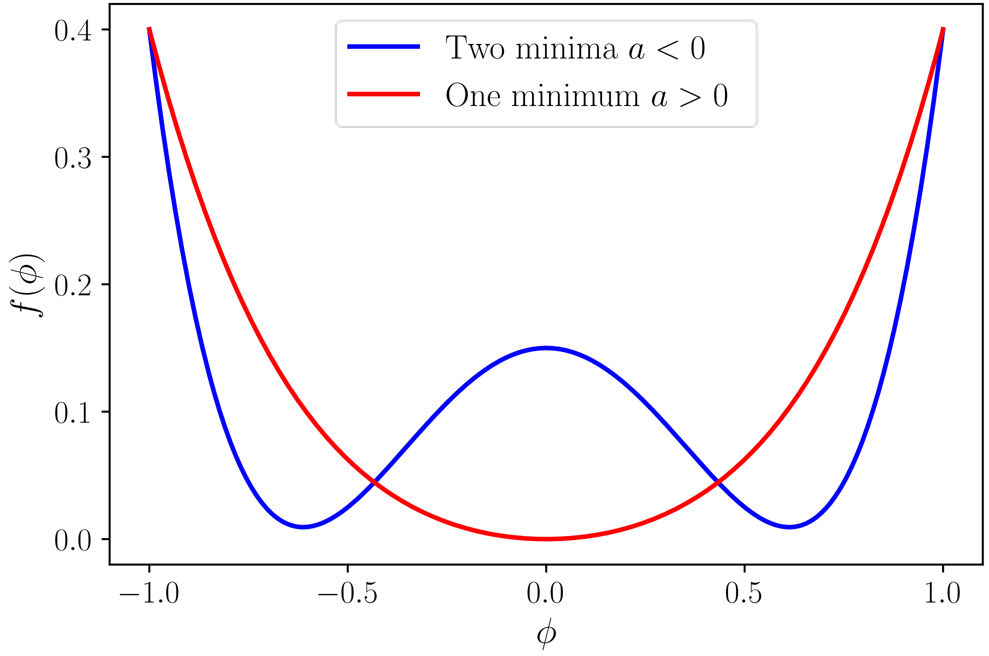
- Measure of the degree of order across the system.
- Eg. mean magnetization, concentration, orientation etc. – call it simply the *order parameter* $\phi(\mathbf{r}, t)$

Classical phase separation – equilibrium vs. active systems

- Animals are not passive – they consume energy to propel themselves. Continuous energy consumption leads to detailed balance breaking.
- Different physical models. No classical energy to minimize

M. Munoz

Modeling phase separation – free energy landscape

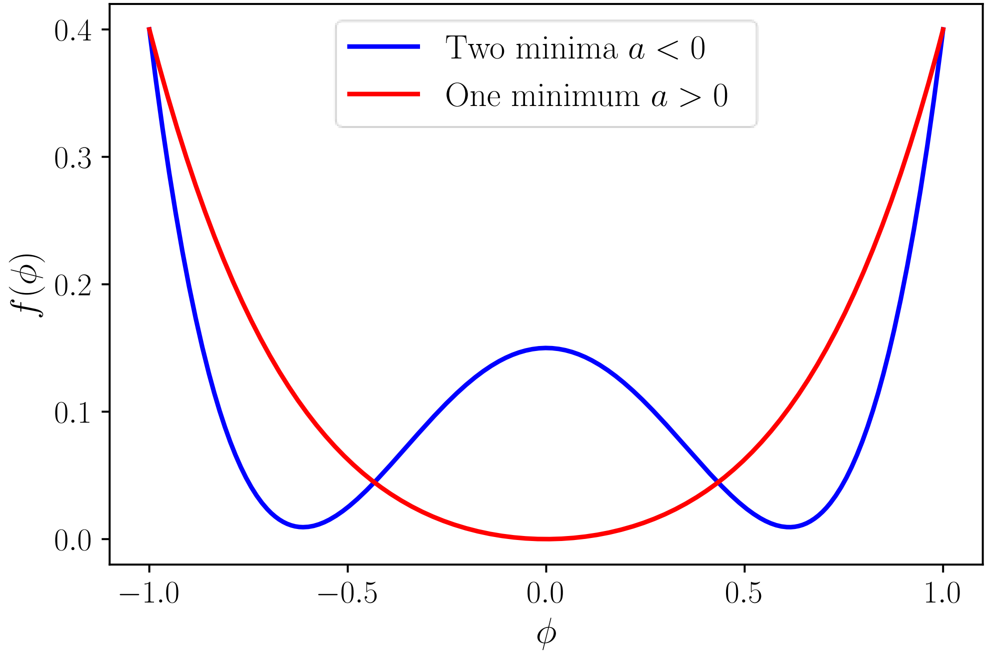


- Approximate the energy landscape using the Taylor expansion:

$$f(\phi) = a_0 + a_1\phi + a_2\phi^2 + \ldots$$

$$f(\phi) = \frac{a}{2}\phi^2 + \frac{b}{4}\phi^4$$

Modeling phase separation – free energy landscape



- System wants to relax into infinitely many small droplets. Want to penalize interfaces in some way – term proportional to the concentration gradient:

$$H = \int \mathrm{dV} \left(\frac{a}{2}\phi^2 + \frac{b}{4}\phi^4 + \frac{\kappa}{2} |\nabla \phi|^2 \right)$$

- Time dependent dynamics – Cahn-Hilliard model (conserved order parameter)

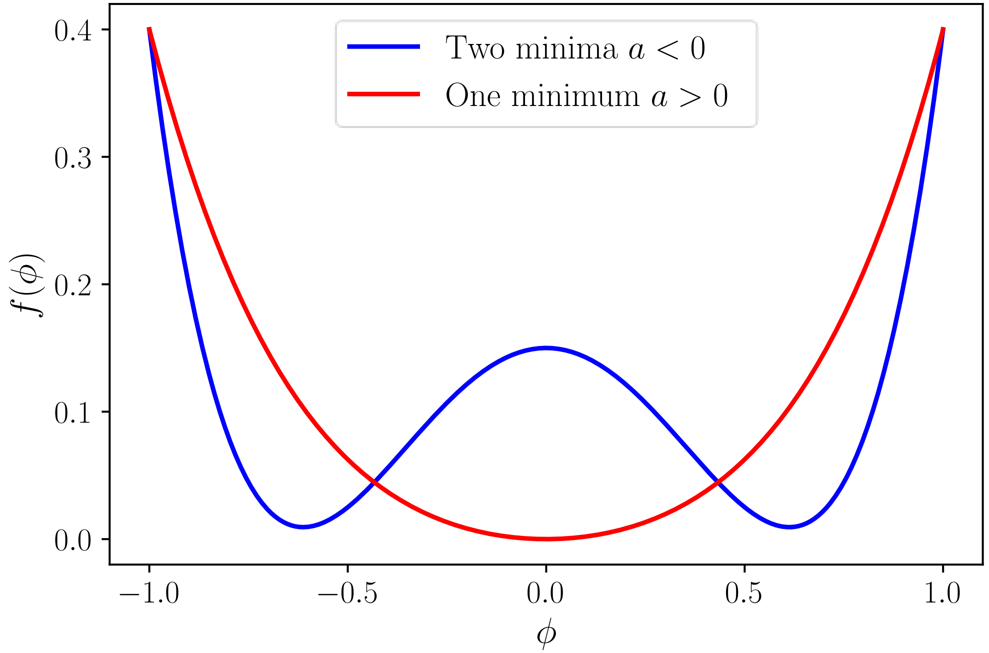
$$\frac{\partial \phi}{\partial t} = M \nabla^2 \mu$$

$$\mu = \frac{\delta F}{\delta \phi} = \frac{\partial f}{\partial \phi} - \kappa \nabla^2 \phi$$

$$\frac{\partial \phi}{\partial t} = M \nabla^2 \left(a\phi + b\phi^3 - \kappa \nabla^2 \phi \right)$$

Modeling phase separation – free energy landscape

- Biologists have been using the analogy of hills and valleys to describe cell division for decades. The free energy landscape gives a rigorous description



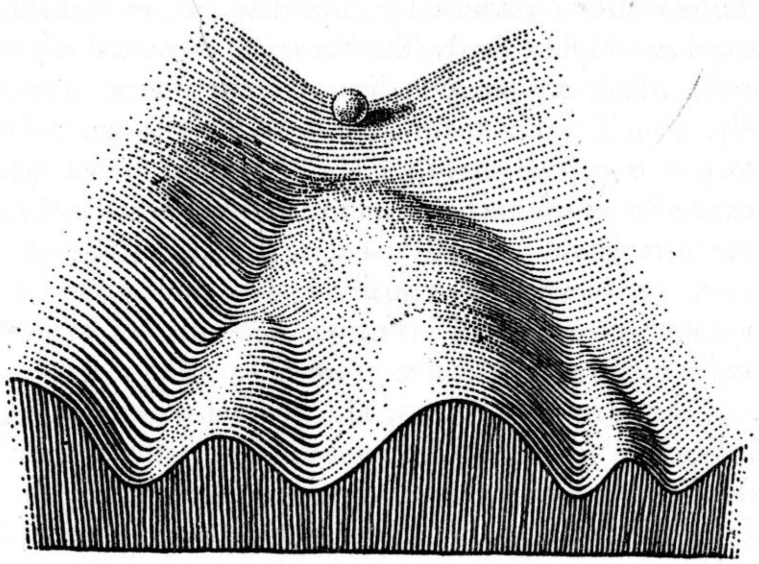
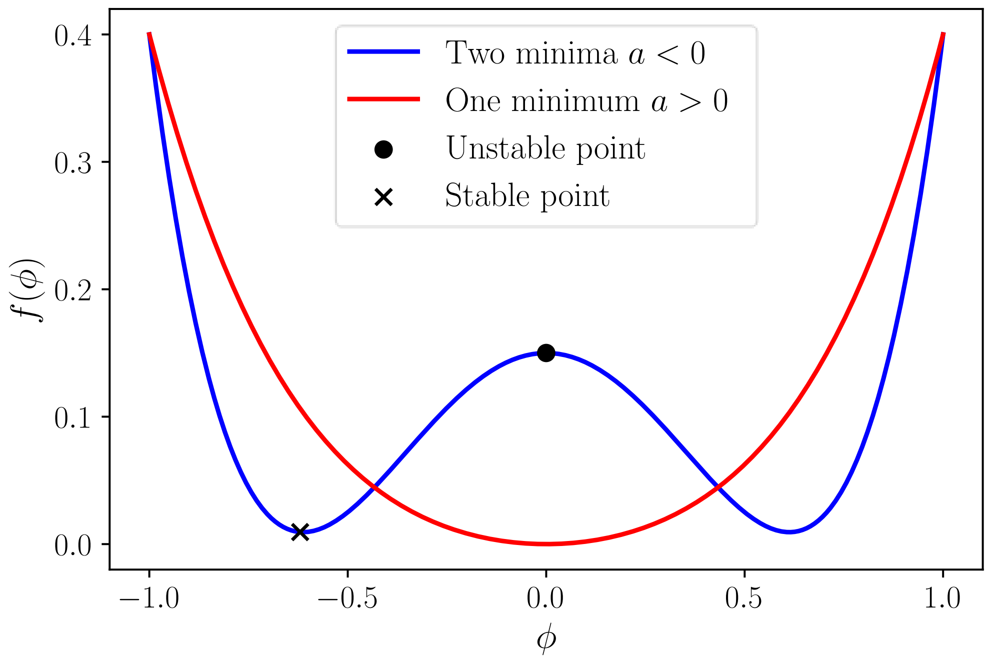


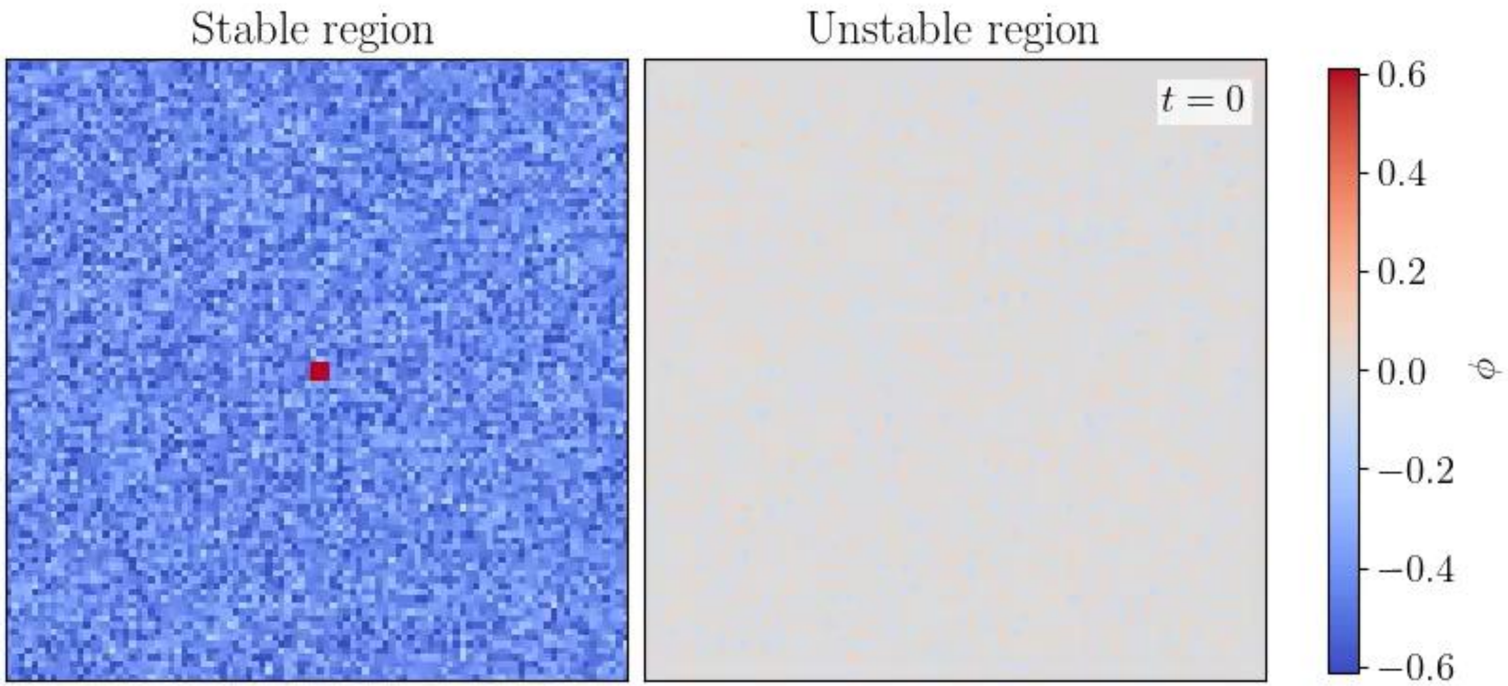
FIGURE 4

Part of an Epigenetic Landscape. The path followed by the ball, as it rolls down towards the spectator, corresponds to the developmental history of a particular part of the egg. There is first an alternative, towards the right or the left. Along the former path, a second alternative is offered; along the path to the left, the main channel continues leftwards, but there is an alternative path which, however, can only be reached over a threshold.

Waddington, 1957

Modeling phase separation – free energy landscape





Simulation: FM

Breakout

1. Why can biological systems benefit from living at the boundary between robustness and adaptability?
2. What's the role of the order parameter? Which order parameters did we introduce?
3. Which collective behaviour in animals do you think can be described as emerging from local interactions? Think of examples

Collective motion of animals

- Collective motion (of a herd, flock, swarm) is an **emergent** collective phase
- Just like molecules, animals use very simple physical interactions to generate global order.

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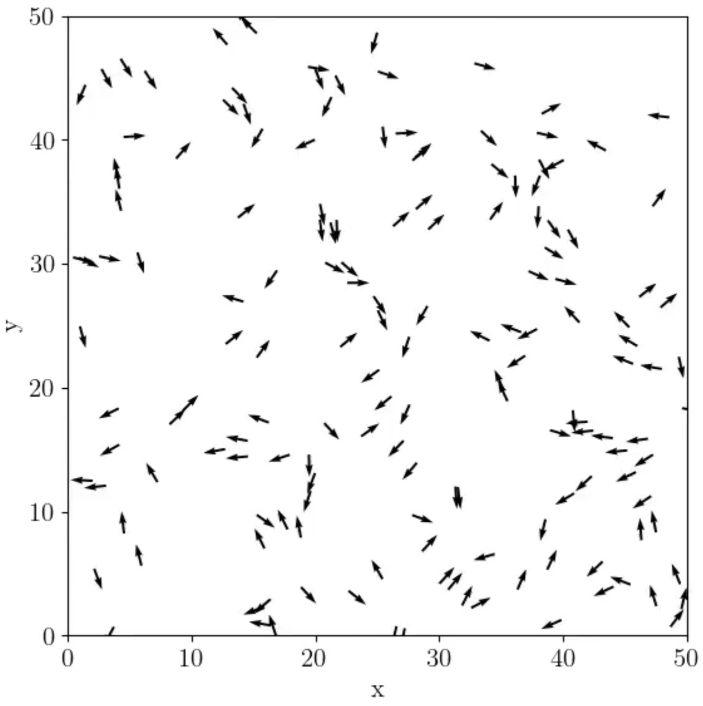
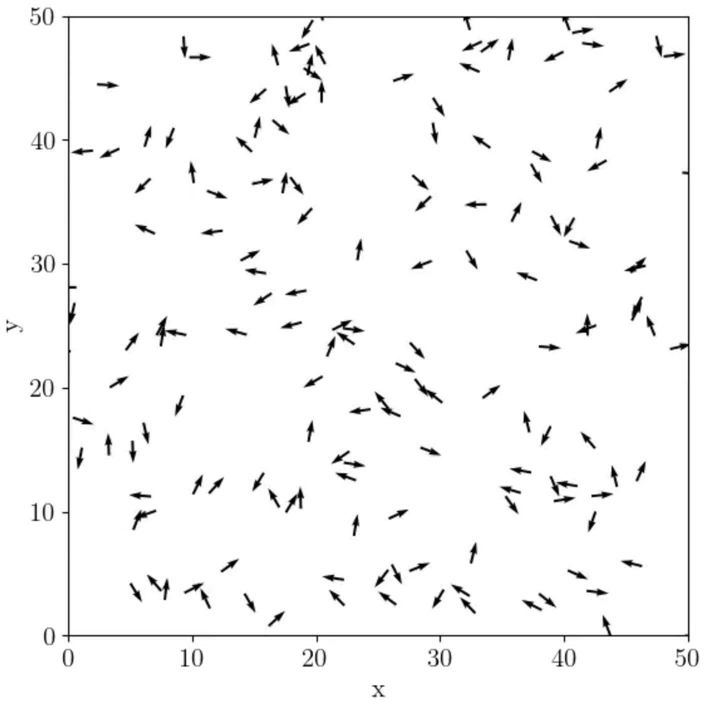
Vicsek model

Governing equations	Ordered phase (noise strength 0.3)	Disordered phase (noise strength 10.0)
$\theta_i\,(t+\Delta t) = \langle\theta_j\rangle_{\lvert\mathbf{r_j}-\mathbf{r_i}\rvert<r} + \eta_i$ $\mathbf{r_i}\,(t+\Delta t) = \mathbf{r_i}\,(t) + v\Delta t \begin{pmatrix}\cos\theta_i \\ \sin\theta_i\end{pmatrix}$		

Vicsek et. al., 1995
Simulations: FM

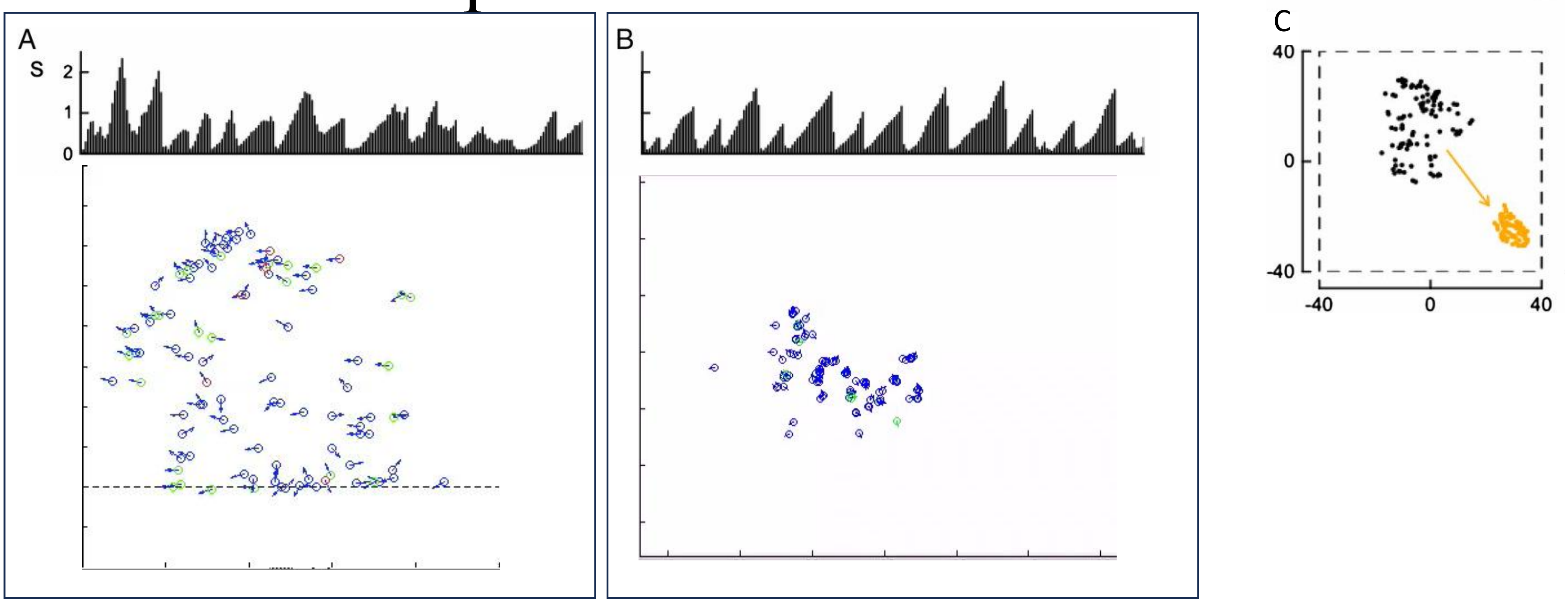
Merino sheep

Youtube, Caters Clips

- Robustness – herbivores need to act as a massive, cohesive group to protect themselves from predators
- Adaptability – if they are too closely packed, they cannot graze effectively and cannot quickly change direction
- Operating at the boundary gives them the best of both worlds

Munoz, 2018

Merino sheep



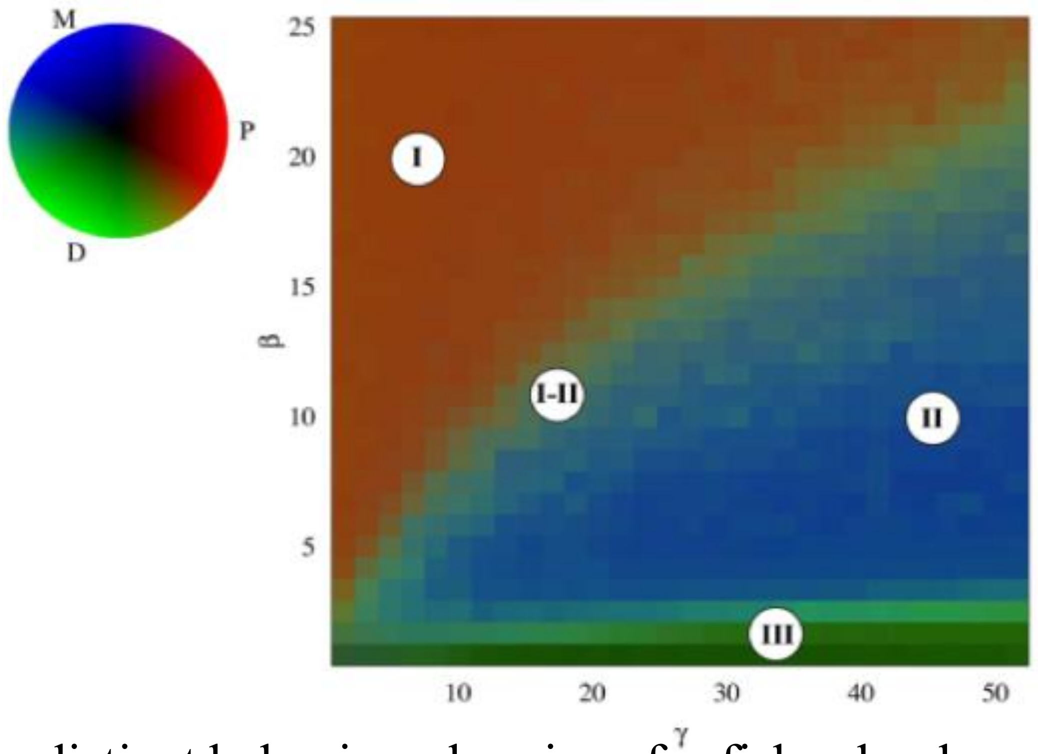
(A) – experimental results: area occupied by sheep S as a function of time
(B) – simulation results for a Vicsek-like model
(C) – packing event

Ginelli et. al., 2015

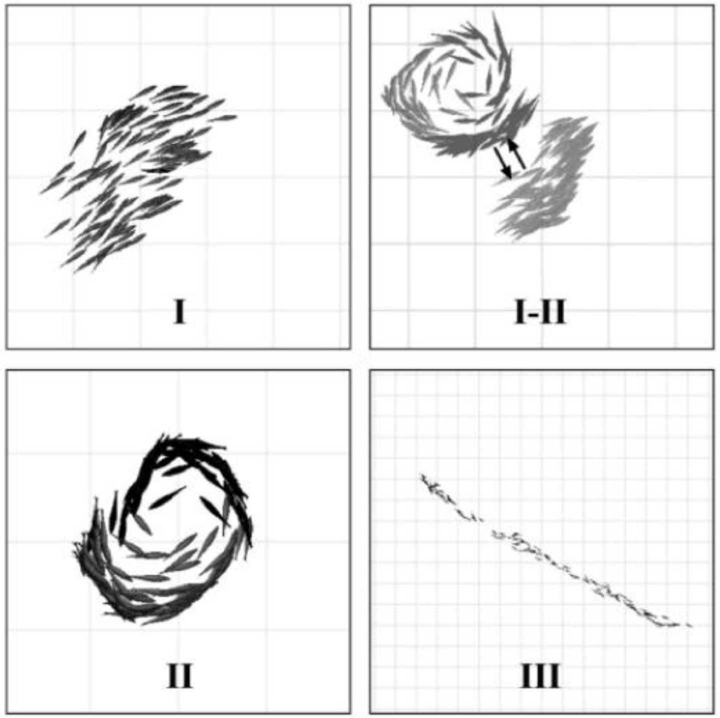
Fish schools exhibit collective behaviour

- Calovi et. al. develop a model for milling and swarming in fish schools. Alignment with neighbours (prefered weights for front alignment) and attraction, with a Weiner-process applied to the angular velocity

A



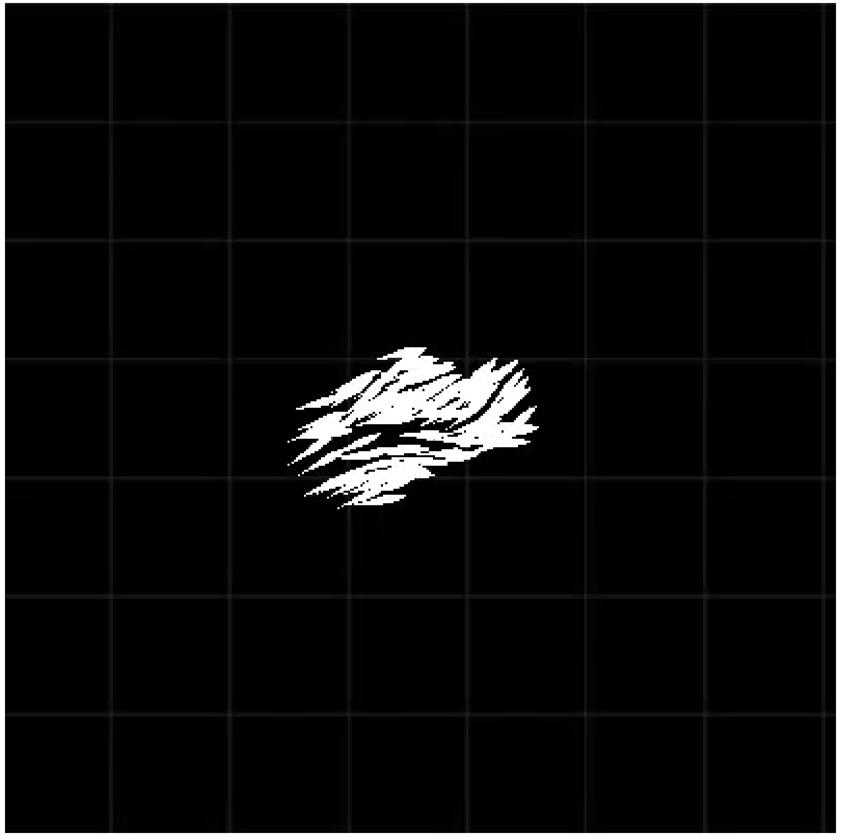
B



Three distinct behavioural regions for fish schools.
(A) shows the phase diagram, and **(B)** the typical behaviours in these phases

Calovi et. al., 2014

Fish schools



Calovi et. al., 2014

Lafoux et. al., 2024

Summary

- Biological systems operate at the boundary between order and disorder
- Classical tools, like the order parameter and Cahn-Hilliard dynamics give us tools to describe these phase transitions
- Active emergence – animals use simple, local interactions to generate global order
- By applying simple active models to the systems, we can describe their behaviour

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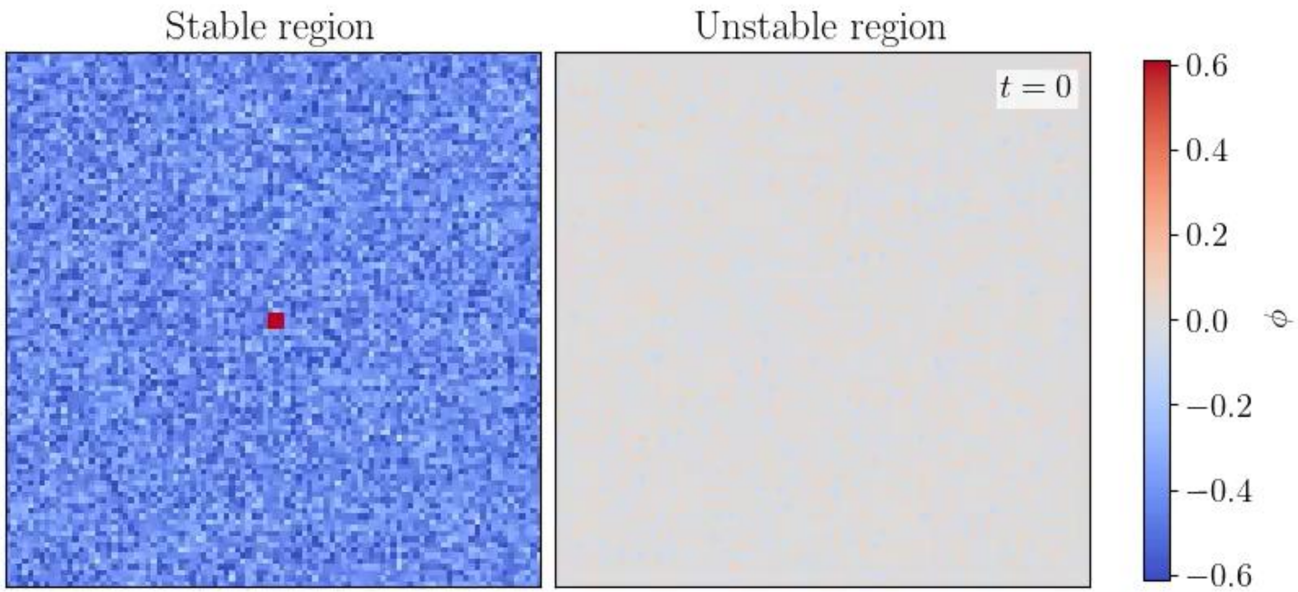

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Youtube, Caters Clips

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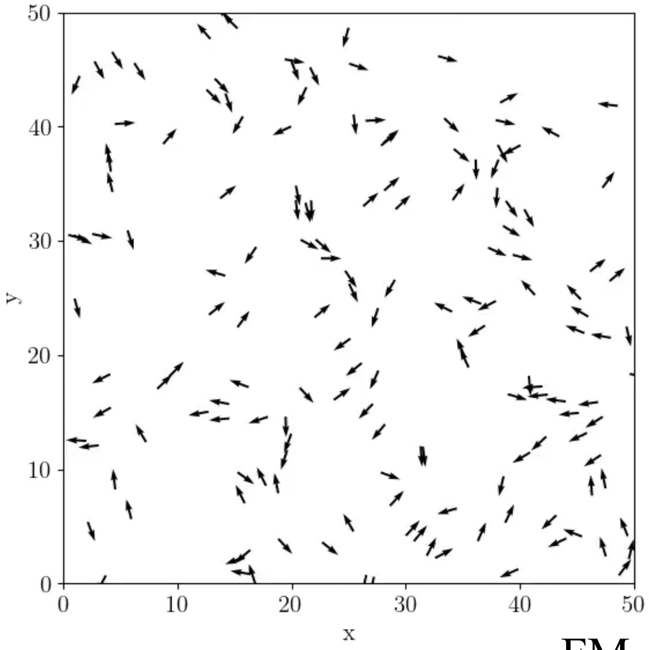
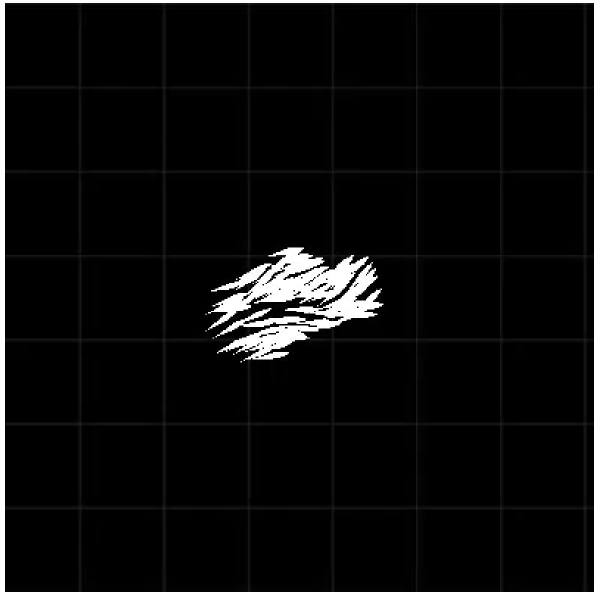


FM



Calovi et. al., 2014

Appendix 1: Binodal and spinodal curves

- Binodal points represent the absolute lowest energy state of the system. Moving from the mixed state into droplets leads to a lower energy, but the surface tension creates an initial energy cost. Need a nucleation point to separate
- Spinodal points are the inflection points. The reward for demixing is immediate. Because the curve points downward, there is **no energy barrier**, so separation happens spontaneously